

Urban Resilience Index (URI) Analysis for the Baldwin Hills & Urban Watersheds Conservancy

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I. Introduction: Baldwin Hills & Urban Watersheds Conservancy



In 2001, the State of California created the Baldwin Hills & Urban Watersheds Conservancy to protect and manage urban parkland in South Los Angeles (Baldwin Hills & Urban Watersheds Conservancy, 2023, p. 4).

This initial territory centered around a two-mile area of The Baldwin Hills parklands and Kenneth Hahn State Recreation Area, which is an area rich in cultural and community value. Historic planning decisions like discriminatory housing tactics and the construction of highways through residential areas, like South Los

Angeles has caused lasting harm to both the local communities and ecosystems. Today, climate change is aggravating these challenges, increasing the risk of flooding, raising the urban heat index, and fueling more frequent wildfires. By creating new protected green spaces and other resilience investments, the Conservancy aims to promote sustainable education using an Urban Resilience Index (URI).

Exhibit 1: demonstrates the previous and new jurisdictional territory of the Baldwin Hills and Urban Watersheds Conservancy

Recognizing South Los Angeles' need for sustainable goals, the California State legislature passed the Baldwin Hills Conservancy and Urban Watershed (BHUWC) Act in 2022, thereby expanding the Conservancy's jurisdiction. This action gave the agency complete authority to coordinate, lead, and manage the parks and open spaces within a 70 mile region that includes the Southern Ballona Creek and Upper Dominguez Watersheds, and extends to other popular Los Angeles cities such as Culver City, Del Aire, Westmont, El Segundo, portions of Manhattan Beach, and more (Baldwin Hills & Urban Watersheds Conservancy, 2023, p. 4). With this large scope, the Conservancy now has the authority and opportunity to combat the long-standing climate and equity issues in many of Los Angeles' main cities.

In addition, in 2024, voters passed Proposition 4, which authorized the state of California to borrow 10 billion dollars to spend on environmental climate projects, also called the California Climate Bond (Cal Matters, n.d.). Of this amount, \$700 million is specifically allocated to California parks and conservancies, which represents a significant investment in the protection and enhancement of the state's open spaces and community green spaces (LAO, 2025). As a

result, now that the Conservancy oversees a significantly larger territory, it may gain access to far greater resources to manage it. To help achieve the Conservancy's goal of improving environmental outcomes in the South Los Angeles area, the Urban Resilience Index (URI) can serve as a critical tool for identifying high-need areas and guiding sustainable investments. By leveraging the URI, stakeholders and community members can better target their efforts to ensure equitable access to climate resilience and environmental benefits across the region.

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The URI is a climate, environmental, and equity mapping tool that layers data to demonstrate how climate change, habitat restoration, and community investments are all connected. The URI Blueprint, developed by the Conservancy that drafts the current URI tool, expresses its mission to better understand the South Los Angeles region while promoting public awareness and providing educational experiences for the community.

The URI is built from layered databases using colored hexagonal data that helps identify clusters or small regions of high need areas (Baldwin Hills & Urban Watersheds Conservancy, 2023, p. 24). Users of the mapping tool can explore existing projects, vulnerabilities to natural disasters, and the ability of their communities to bounce back from difficult natural disasters. For example, the Parks and Open Space Access resilience category will help a user identify the lack of green and open spaces in their neighborhood. This enables community members and stakeholders to visualize current initiatives across the region while also identifying opportunities for new projects and targeted funding that advance the Conservancy's mission to serve the broader Los Angeles area.

II. Resilience Categories

Stated in the URI Blueprint, resilience is defined as the capacity of individuals, communities, institutions, businesses, and systems within a city to survive and thrive no matter what kinds of chronic stresses or acute shocks they experience (Baldwin Hills & Urban Watersheds Conservancy, 2023, p. 20). The URI is broken up into five resilience categories: Community Empowerment, Water Management and Protection, Restoring Degraded Lands and Increasing Biodiversity, Mitigating Impacts from Extreme Heat and Fire, and Park and Open Space Access (Baldwin Hills & Urban Watersheds Conservancy, 2023, p. 21).

The current five categories, addressed in the URI Blueprint, describe the following resilience types. Firstly, the Community Empowerment Resilience Category aims to support social and economic justice through workforce development, improved housing, and expanded local services. Secondly, Water Management and Protection addresses water-related challenges like flooding or water quality through strategies such as stormwater capture or water resistance. Thirdly, Restoring Degraded Lands and Increasing Biodiversity targets the restoration of plants, green spaces, and air quality improvements. Next, Mitigating Impacts from Extreme Heat and Fire prioritizes climate resilience with shade infrastructure and fire prevention efforts. Lastly, Park and Open Space Access seeks to expand green space availability by improving transportation and enhancing recreational opportunities. Together, these evolving categories provide a framework for assessing the urban resilience of the Conservancy's territory, as represented in the Urban Resilience Index.

a. Resilience Category Layer Analysis

The group evaluated the original five resilience categories using the evaluation criteria below (Exhibit 2) and provided a score for each criterion (Exhibit 3). A score of 1 (low vulnerability) is the lowest possible score, whereas a score of 3 (high vulnerability) is the highest.

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Exhibit 2: Evaluation Criterion for different resilience categories.

Evaluation Criteria	Definition
Accessibility	How self-explanatory is this indicator (1-3)?
Transparency	Is the data listed online (1)? Can you download it (1)? Has it been updated in the last 5 years (1)?
Comprehensiveness	How well do you think this indicator represents the resilience category (1-3)?

Exhibit 3: Scoring of each resilience category.

Resilience Categories	Indicator	Accessibility	Transparency	Comprehensiveness	Average Score
Community Empowerment	Social Vulnerability Index	2.0	3.0	3.0	2.7
Water Management and Protection	FEMA Flood Hazard Risk	2.0	3.0	2.4	2.5
Restoring Degraded Lands and Increasing Biodiversity	Diesel Particulate Matter	2.0	2.25	1.25	1.8
Mitigating Impacts from Extreme Heat and Fire	Heat Health Action Index	2.0	2.0	1.75	1.9

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Park and Open Space Access	Percent of people living within a walkable distance (half-mile) of a park, beach, or open space. (WIN: Well-being in the Nation).	3	2.5	2.75	2.7
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b. Resilience Category Score Feedback

“Community Empowerment,” measured using the Social Vulnerability Index, received an average score of 2.7 based on accessibility, transparency, and comprehensiveness. While social vulnerability includes a large variety of components like income level, disability, status, housing, and language barriers that help determine social vulnerability, it can be difficult to understand at first. The index explains each indicator in detail and effectively identifies population characteristics that increase vulnerability, which is transparent (3.0) to any user. However, the accessibility (2.0) score is slightly lower, as the index can be difficult to interpret without prior knowledge or experience with data or the definition of social vulnerability. Additionally, the volume of data can be overwhelming for users. Therefore, a simplified summary or visual representation would greatly improve usability for community members and non-experts.

“Water Management and Protection” received a strong average score of 2.5, indicating that FEMA Flood Hazard Risk is a valuable and relevant indicator for assessing vulnerability to flooding. However, while it effectively captures one important aspect of water-related risk, it does not address the full spectrum of water management challenges, such as drought resilience, water quality, groundwater sustainability, and stormwater infrastructure. Adding complementary indicators that reflect these additional dimensions would provide a more holistic understanding of water management and strengthen the category overall.

“Restoring Degraded Lands and Increasing Biodiversity” received an overall score of 1.8, largely due to the use of “Diesel Particulate Matter” as its primary indicator—an indicator that does not directly capture the ecological health, land restoration efforts, or species diversity typically associated with this category. While diesel particulate matter is a valid environmental health concern, it is more closely related to air quality and human exposure to pollution than to habitat recovery or biodiversity outcomes. As a result, it fails to reflect the core goals of this category, leading to a lower and potentially misleading score. A more relevant set of indicators—such as native vegetation cover, habitat connectivity, or presence of key species—would provide a more accurate assessment.

“Mitigating Impacts from Extreme Heat and Fire” received an overall score of 1.9, highlighting the limitations of using the Heat Health Action Index as the sole indicator. While this index provides valuable insight into heat-related health preparedness, it does not address wildfire risk, which is a distinct and increasingly urgent concern, especially in Southern California. As such, the indicator fails to fully represent the dual focus of the category. To improve accuracy, the group recommended separating the category into two: “Mitigate Impacts from Extreme Heat” and “Prevent Wildfire,” which was already implemented by the Conservancy. For the wildfire component, risk could be more appropriately assessed using data from CAL FIRE’s Fire Hazard Severity Zones, which offer spatial information on areas most vulnerable to wildfire based on factors like vegetation, topography, and historical fire patterns.

“Park and Open Space Access” received an average score of 2.7 across the three categories. The indicator for this category was the percent of people living within a walkable distance (a half-mile) of a park, beach, or open space greater than one acre. The accessibility score was a 3 since the data was specific and well defined in what it was measuring.

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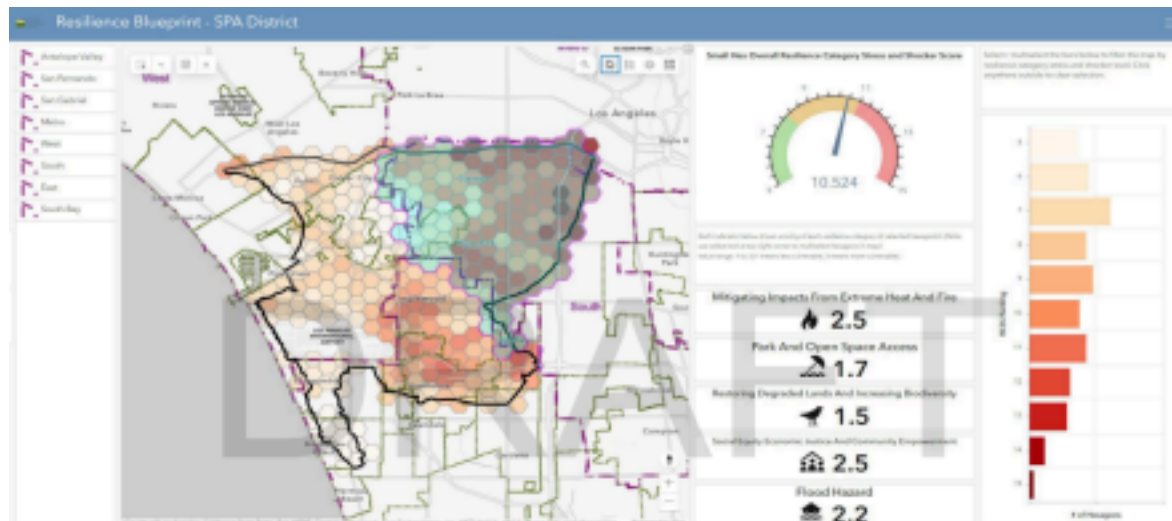
Additionally, this indicator scored a 2.75 in the comprehensiveness category since the data reflects the most direct definition of park and open space access. It does not cover all of the park and open space access since there may be more opportunities within a slightly larger distance that can be easily accessible through transportation. The transparency score averaged 2.5 since the data had not been updated since 2017.

The conservancy, developing these five categories through research, analysis, agency plans, community organizations, and literature reviews, has realized that these categories will still evolve. Therefore, as the Conservancy identifies and explores other indices, it has become clear that expanding the index to eight categories will allow for a more comprehensive and digestible map. The eight categories, descriptions, actions, and sub-categories are still evolving and are continuously being developed.

III. The Existing URI Tool

The URI provides a composite score meant to represent the vulnerability of a region or system based on five core categories, called resilience categories. These categories represent a key dimension that influences a system's ability to respond to and recover from stressors, with each category being scored on accessibility, transparency, and comprehensiveness. Despite the name, higher URI scores indicate greater vulnerability. The index ranges from five to 15, with each of the five categories scored from one (low vulnerability) to three (high vulnerability). A total score of 15 indicated maximum vulnerability added across all categories, while five means low vulnerability. Therefore, this makes the URI a quick visual and numerical summary of an area’s resilience.

Exhibit 4: URI and Breakdown of Resilience Categories



For example, this image demonstrates an overall vulnerability of 10.524. This number is calculated by calculating the sum of each of the five resilience categories. Overall, there is a mid-to-high vulnerability in the area.

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By ensuring the URI is accessible, intuitive, and inclusive of diverse perspectives, it can empower planners, community members, and agencies to create a safe, equitable, and environmentally sustainable future for Baldwin. Therefore, the URI is essential to Baldwin Hills in strategic sustainable planning, data-driven decision making, multi-benefit project designs, and community engagement and education. This paper aims to make recommendations for how the URI can be improved by examining other tools similar to the URI, by comparing and contrasting ways in which the URI can be enhanced to increase community comprehension, design, and features.

IV. Cross-Comparison: Looking at Other GIS Mapping Tools

The group analyzed three other GIS-based mapping tools to understand the strengths and limitations of the Urban Resilience Index (URI) tool. This comparative analysis allowed us to evaluate different approaches to data visualization, usability, and the types of indicators included. By identifying features that worked well in other tools, we were able to generate concrete recommendations for improving the URI tool's functionality and relevance. This included suggestions for new datasets, interface improvements, and potential metrics to add or refine. Below is a breakdown of each tool we reviewed and the insights they provided.

a. Tool: Resilient Land Mapping Tool

Creator: Nature Conservancy – Resilient Land Mapping Tool

Overview of the Tool:

The Resilient Land Mapping Tool was created by the Nature Conservancy to identify

areas with the physical and ecological characteristics necessary to sustain biodiversity and natural systems under climate change.

The Resilient Land Mapping Tool is an interactive GIS-based platform that helps users locate landscapes with strong potential to support biodiversity over time. It evaluates ecological resilience through three key indicators: landscape diversity, local connectedness, and climate flow. Each of these layers reflects natural features that promote species movement, habitat integrity, and climate adaptation. The map interface allows users to explore these indicators at regional and national scales, with options to create or upload polygons to analyze a specific area. Its primary audience includes conservation organizations, land-use planners, and policymakers engaged in long-term environmental and climate planning. This tool is particularly valuable for identifying areas that can serve as climate refugia for species or that should be prioritized for conservation investment. The tool is not intended to address urban or community-level resilience like the URI does. It lacks data on social vulnerability, infrastructure, and environmental hazards like heat or wildfire, which limits its relevance for broader climate adaptation planning in urban settings.

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Cross Comparison: Resilience Category (RC) Representation – RC 3: Increase Biodiversity

The Resilient Land Mapping Tool most directly relates to RC 3: Increase Biodiversity. This is the core of the tool's design. Through its use of geophysical settings, intact habitats, and ecological connectivity data, it highlights landscapes that offer the best chance for biodiversity to persist despite climate change. The "Recognized Biodiversity" and "Resilient Biodiversity" layers of the tool were assembled using information on places recognized for their biodiversity value (rare species, intact habitat, or exemplary natural communities) in separate studies. The "Landscape Diversity" layer refers to the microhabitats and climatic gradients available in the immediate neighborhood surrounding any 30-m cell of land. The persistence of species in an area increases in landscapes with a wide variety of microclimates created by the topography (topo-climates), elevation, and hydrology, and using that data set could benefit the URI as well.

While RC 3 is thoroughly covered, most of the other Resilience Categories are **not** addressed by this tool. It does not include data on social vulnerability (RC 1), urban tree canopy (RC 4), heat or wildfire risk (RCs 7 and 8), or degraded land (RC 5). Some layers, like "climate flow," may indirectly relate to water movement (RC 6), but there are no explicit indicators for flood zones or water resource protection. The absence of these categories limits the tool's use for urban planners or equity-focused initiatives. However, for users focused on ecological integrity and long-term biodiversity conservation, the tool is exceptionally well-suited.

User Interface & Experience Analysis:

The user interface of the Resilient Land Mapping Tool is streamlined and accessible for those familiar with spatial data. The layout is clean, with a left-hand sidebar allowing users to toggle different layers like "Terrestrial Resilience," "Recognized Biodiversity," and "Climate Flow." The map uses a consistent color gradient to represent data values, helping users visually

distinguish areas of high and low ecological resilience. Creating a polygon allows the user to analyze a very specific area and determine how resilient it is.

Despite its strengths, the tool could be improved by incorporating more user support features. There is a short introductory walkthrough, but it is not thorough enough. Including a more detailed walkthrough or glossary for new users, and some of the scientific terminology (e.g., “climate flow” or “local connectedness”) might help give some background knowledge. This will especially be useful for the URI, as it will be used by community members and needs to have a user-friendly interface. Additionally, the Resilient Land Mapping Tool provides explanations of what each layer is, which could benefit the URI.

b. Tool: Connecticut Environmental Justice Mapping Tool

Creator: University of Connecticut – Connecticut Institute for Resilience and Climate Adaptation (UConn CIRCA)
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Overview of the Tool:

The Connecticut Environmental Justice Mapping Tool is an interactive GIS-based platform developed by the Connecticut Institute for Resilience and Climate Adaptation (CIRCA). It was designed to help visualize and assess environmental burdens and social vulnerabilities across the state. This tool also supports community-based environmental justice (EJ) work by providing accessible, data-rich insights into cumulative risks faced by historically marginalized communities.

The tool organizes data into three major categories: Potential Pollution Sources, Pollution Exposure, and Health Sensitivity. Users can explore layers that include proximity to brownfield sites, landfills, incinerators, hazardous chemical facilities, and wastewater discharge locations. It also includes air quality indicators such as diesel particulate matter, ozone levels, PM2.5, and air toxics. The Health Sensitivity category underscores populations at greater health risk, including those with high asthma emergency visit rates.

Intended for use by policymakers, planners, researchers, advocates, and community members, the platform supports initiatives that address environmental and public health disparities. While powerful, the tool may be overwhelming for new users due to the large volume of indicators.

However, once users become familiar with the interface, it reveals clear patterns of environmental injustice, with many high-impact indicators clustering in low-income areas.

Cross Comparison: Resilience Category (RC) Representation

The Connecticut EJ Mapping Tool intersects with several resilience categories in the Urban Resilience Index (URI) framework. It aligns most directly with RC 1: Community Empowerment, as the tool provides critical insight into how pollution and health burdens disproportionately affect low-income communities and communities of color. It allows

community members and advocates to identify and raise awareness about these inequities using concrete, place-based data.

The tool also supports RC 3: Restoring Degraded Lands. Layers such as Brownfield Sites, Lead Risk in Housing, Underground Storage Tanks, and Potentially Contaminated Sites illustrate where land remediation may be needed to protect public health and restore ecological function. RC 4: Mitigating Extreme Heat and Fire is addressed through the Urban Heat Island layer, which identifies urban areas with elevated temperatures, often overlapping with regions that lack green space or tree canopy. Additionally, RC 6: Water Management and Protection is represented through indicators such as Wastewater Discharge and Impervious Surfaces. These help identify vulnerabilities in water infrastructure and highlight areas more prone to flooding or water contamination.

While the tool does not provide direct data on biodiversity (as represented in RC 3 of the Resilient Land Mapping Tool) or park access (RC 5 in URI), it offers a broad, intersectional view

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of human and environmental health risks. This supports community-driven resilience planning and enhances the framework established by the Urban Resilience Index.

User Interface & Experience Analysis

The user interface of the Connecticut Environmental Justice Mapping Tool is well-organized, although it may feel overwhelming for users without a background in GIS or environmental data. The platform utilizes a side panel to toggle between dozens of layers across categories like Pollution Sources, Pollution Exposure, and Health Sensitivity. Each layer is accompanied by a brief description, and data is presented using a consistent 0–10 scoring scale with intuitive color gradients. This visual framework enables users to quickly identify which areas are most burdened by environmental hazards.

For more advanced users, such as researchers or planners, the tool is highly functional. It includes source links, update timelines, and download options for most datasets. However, it currently lacks an onboarding process for new users. A more detailed walkthrough, glossary of key terms, or example applications would significantly improve accessibility for students, advocates, and community members unfamiliar with environmental data tools.

Implications for Improving the Urban Resilience Index (URI)

The Connecticut Environmental Justice Mapping Tool provides several strong design features and methodological approaches that could inform improvements to the Urban Resilience Index. For example, the tool's 0–10 scoring system and visual layering of environmental indicators could be adopted within the URI to better communicate the severity of shocks and stresses across regions. Its structured indicator categories could also enhance URI users' ability to interpret and engage with complex resilience data.

Additionally, features such as detailed source documentation, layer descriptions, and

public health indicators—including asthma emergency visit rates and PM2.5 concentrations—could expand the URI’s analytical scope, particularly under RC 1 (Community Empowerment) and RC 4 (Mitigating Extreme Heat and Fire). Overall, the Connecticut tool offers a clear model of how environmental data can be made accessible, transparent, and actionable, and its strengths in design and usability present valuable lessons for future updates to the URI platform.

c. Tool: The Living Infrastructure Field Kit

Creator: Accelerate Resilience L.A. (ARLA), Spherical

Overview of Tool:

The Living Infrastructure Fieldkit is a free online tool meant to support LA residents, non-profit organizations, and other agencies to plan and implement living infrastructure projects. Living infrastructure refers to natural systems that provide services for both the environment and

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people (e.g, vegetable gardens, urban forests, stormwater sites)The creators of the tool include Accelerate Resilience L.A. (ARLA), Spherical, and various organizations across L.A. County, with LA residents serving as the intended audience of the tool.

The goals of The Field Kit can be summarized in four categories. Category I: Site Profiles, Category II: Understanding Places Using Data, Category II: Community Engagement and Collaboration, Category III: Project Design. The field kit has the option to create a site or click a pre-generated one, then access over 65 layers of data to further understand these sites (e.g, vacant lots, fire, flood risk, heatwave temperature). The next step would be community engagement, where one can invite other stakeholders to be part of the sketch and document ideas from everyone. The final step is Project Design, where one can apply remedies to the site with the estimated benefits and costs. The tool provides valuable educational materials with remedy descriptions, cost-benefit analysis, and remedies. For instance, users can create a site, click on the “Food Access” layer, which will show areas with limited access to healthy and affordable food based on distance to food stores, income levels, and other factors. The field kit explains the methodology, what “low food access” looks like, and what “high food access” means. One of Fieldkit’s strengths is its ability to add the recommended remedies to the site and visually see how the remedies changed the scores of the site, while giving one a breakdown of the costs associated with those remedies, both the creation and the annual care costs.

Cross Comparison: Resilience Category (RC) Representation

The datasets in the Living Infrastructure Toolkit intersect with various resilience categories in the URI; there are over 65 layers to the field kit, allowing the tool to interact with the 7 resilience categories. This includes R1 Community Empowerment, R2 Water Management & Protection, R3 Restoring Degraded Lands and Increasing Biodiversity, R4 Mitigating Impacts from Extreme Heat & Fire, and R5 Parks and Open Space.

The field kit covers R1 Community Empowerment, as the Field Kit was specifically

made for community members to create their own living infrastructure projects. For instance, the layer pack that creates “Sense of Place” includes the following layers: Historical Landmarks (Civic Art, Murals). The Fieldkit describes a sense of place as “increasing community cohesion, stemming from social, environmental, structural, and personal factors,” which is a similar definition of community empowerment to the URI. The URI can use the field kit’s Sense of Place layer pack from the following datasets: National Register of Historic Places, Mural Map LA, and LA County Civic Art Collective. With R2 Water Management & Protection, many layers from the Field Kit that cover this resilience category, including Groundwater Basins, Impaired Waterbodies, Runoff Contribution, SCWP Watershed Areas, and Sand Subwatershed. R3 is covered in the field kit with Significant Ecological Areas, and similar to the Sense of Place pack, there is a Habitat pack that includes: Leaf Area Index, Significant Ecological Areas, and Waterbodies.

User Interface & Experience Analysis:

The Field Kit has a homepage that explains the Field Kit, its purpose, and gives insight into how to use the tool. This is essential information for a user, as one should familiarize oneself

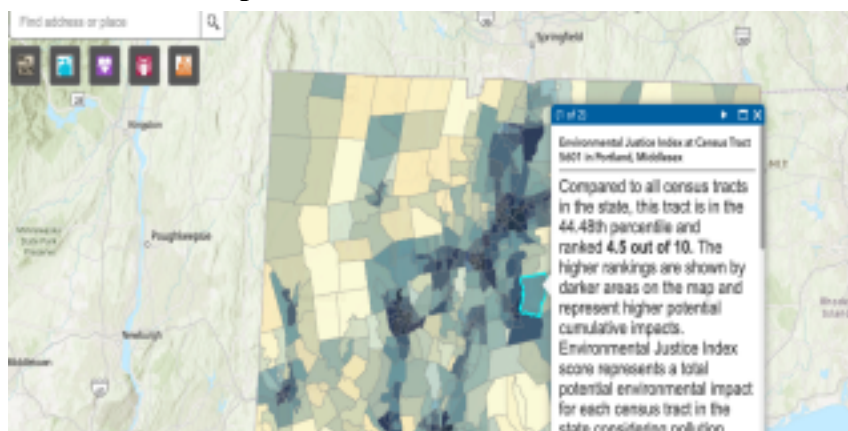
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with how the tool works before creating a site and analyzing the layers. In addition to the “sites” automatically created by the Fieldkit, one can create their own site and look at how the layers affect it. The field kit was available in two languages, English and Spanish; however, the Spanish version was limited. There were certain sections of the tool that were not translated into Spanish, and none of the videos were translated into Spanish. These were short informational videos, less than 3 minutes, explaining how the various “remedies” can play out. These “remedies” were essentially infrastructure that can change the score of the layer, for example, how a Bikeway can improve the mobility, air quality, public health, etc. One also has the option to directly add these remedies to their site and write notes next to them; the ability to add markers or some type of notes could be beneficial to the URI.

V. Recommendations

After analyzing the three other GIS mapping tools, we filtered the strengths and weaknesses of each tool and how they applied to the URI and its resilience categories. From there, we have adapted these recommendations to promote user comprehension and accessibility. The recommendations are divided into four categories: data improvements, interface enhancements, user accessibility, and optional enhancements. Each category has several suggestions that target specific challenges and areas of improvement in the URI tool.

A. Data Improvements



Starting with the data improvements, these recommendations aim to guarantee that the data used is relevant, accurate, and

comprehensible to the public. One of the main recommendations is to standardize and update the sources. The Connecticut EJM Tool was inconsistent in how it presented data across different areas. When clicking on various areas, some pop-ups displayed bar charts while others showed definitions with external links, making it difficult to navigate and compare

Exhibit 5: Connecticut Environmental Mapping Tool

information uniformly. This discrepancy between the data and how it is represented makes it difficult for users to understand. The URI should include datasets that have been updated within the last 5 years. Using a consistent measurement across all of the data within a category is valuable for making appropriate comparisons and can help understand the context of the numbers. Incorporating historical data can also help by displaying long-term resilience trends

within each resilience category. Older data can serve as a benchmark to compare current data and create connections that help users understand how their community has changed.



Exhibit 6: Living Infrastructure Field Kit historical wildfire layer

score and define what the map shows. Focusing on one term will avoid confusion and discrepancies that occurred when using both. If users wish to research the dataset, there should be a link or embedded area within the description where they can learn more about the topic in the tool and through other sources. This helps users further comprehend the data and answer any questions that may arise while using the tool.

The URI tool's effectiveness is reliant on having current data that is regularly updated, with transparent source material to provide an accurate reflection of the community's resilience. One recommendation was to ensure that all the mapped areas have the same amount and type of information available. This creates an even understanding of the resilience subcategory across communities, and users can learn from the conditions of nearby areas. Prioritizing high-impact

Clear definitions and intuitive displays improve how users interpret resilience categories. Data sources should be linked with information on when it was last updated to ensure they are accurate and applicable for the category. Clearly labelling the resilience categories and including a short description of how each subcategory affects the resilience or vulnerability of the community makes the data understandable and relevant to mapping. The tool needs to distinguish and use either resilience or vulnerability to

and the most relevant data from each resilience category can prevent clutter within the tool and unnecessary information from influencing the urban resilience index. By filtering through the data and how impactful it is on the resiliency index score, some datasets can be removed that are not as accurate or reflective of urban resilience. That data, such as Air Quality that was originally used as a metric for biodiversity, can be helpful as background information and can be included within the descriptions and linked or embedded, but does not need to be included in the layers within the tool. An optional feature to consider is to include local input for certain data categories, such as temperature. The Living Infrastructure Field Kit has an option for users to make observations to document the experience of community members and their opinions on the current conditions.

Exhibit 8: Recommend Alternative Datasets for Each Resilience Indicator

Reliance Category	Original Indicator	Recommended Alternative Dataset/Tool	Relevant Layers or Features
Community Empowerment	Social Vulnerability Index (SVI)	Living Infrastructure	Sense of Place, Disadvantaged Communities, Community Gardens, Community Organizations, Walkable

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			Community Spaces
Water Management and Protection	FEMA Flood Hazard Risk	Living Infrastructure	Flood Risk, Flood Safety, Groundwater Basins, Hydrologic Watersheds, Impaired Waterbodies, Runoff Contribution
Restoring Degraded Lands & Increasing Biodiversity	Diesel Particulate Matter	Living Infrastructure & Resilient Land Mapping Tool & Connecticut Environmental Land Mapping Tool	Pollution-Related Layers, Air Polluting Facilities, Air Quality, CO2 Reduction, CES Score, GHG Sites, Pollution Sources, Clean-up Sites, Pollution Layers, Recognized Biodiversity
Mitigating Impacts from Extreme Heat and Fire	Heat Health Action Index	Living Infrastructure and Connecticut Environmental Land Mapping Tool	Urban Heat Islands, Heatwave Temperature, Cooling Centers,

			Fire Hazard
Parks and Open Space Access	% within ½ mile of a park (WIN)	Living Infrastructure & Resilient Land Mapping Tool	Park and Open Space Maps, Amenities Info, 10-minute Walkability, Density Isochrones, Leaf Area Index, Trees, Landscape Diversity, Carbon Layers

B. Interface Enhancements

The aesthetic design of a mapping tool is crucial in determining how a user will interact with the tool. Adding a welcome page with a tutorial that explains the tool's purpose, the data sources, and the basics of navigating the tool's interface can ensure that users easily understand the tool.

Another crucial feature is incorporating ways to simplify or condense the information of the categories and subcategories, such as relevant filters and search options. Although each of the three tools had several layers and subcategories, there were ways to turn on and off certain segments to focus on one dataset at a time. This can help make all the information within the tool more accessible by limiting the amount of information displayed at once, and help users find the information they are looking for more easily. Both the Living Infrastructure Tool Kit and the Resilient Land Mapping Tool let users draw a polygon to focus on a certain region and how the data applies only to that area. Tool designers should also make contextually appropriate visual design choices for the subcategories, such as picking colors and icons that reflect the definitions and meaning of the data, like in the Living Infrastructure tool. Another interesting feature incorporated into the Living Infrastructure Field Kit was short and informal explanation videos within the expanded definitions of the data categories. This helped show specific local examples within the community of where they could go and experience what this resilience indicator is in

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real life, not just as words or datasets. Grounding the data with human experiences makes the tool much more valuable in impacting how the community addresses more vulnerable areas.

C. User Accessibility

Incorporating certain features that make the tool more accessible can allow it to reach more residents. One interesting feature is incorporating multilingual options. Community members who may not know English well, or at all, can understand the data more clearly through their native language. Additionally, adding an offline option for accessing the data, such as PDF printable summaries, tables, maps, and charts, can be useful for those who do not have access to technology or want to use it for other purposes beyond directly within the tool. The Living Infrastructure Tool Kit allows users to customize accessibility adjustments that can enhance website visuals for vision impairment, provide a screen reader, scale content, and more. Integrating a tool that can automatically adjust the website's interface for user accessibility would

be an ideal way to address accessibility across several categories.

D. Future Development

Currently, the tool is designed to analyze and map out areas of vulnerability within the Baldwin Hills Conservancy. Since the goal is to use projects to improve these areas, the tool can incorporate features, such as the “remedies” tool in the Living Infrastructure Field Kit, that allow visitors to place solutions within a certain mapped area and see the cost, benefits, and upkeep associated with incorporating that solution into the community. It also shows how it affects the existing conditions and the different layers. Incorporating a similar feature can help test where smaller projects can have the greatest benefit. Testing these solutions can become a collaborative process by incorporating public suggestions and comments available to users, either on an account basis or a public forum, to generate useful feedback before completing a project to ensure that it has support from the community.

Exhibit 7: Recommendation Summary

Category	Recommendations
Data Improvements	- Standardized and updated sources - Include historical trends to display long-term resilience - Link to the data source and when it was last updated - - Clearly labeled resilience categories, including a short description of their relation to urban resilience - Embed or link to resources where users can further research or learn about the data - Audit datasets to ensure all areas have the same information available - Prioritize high-impact / most relevant data per category - Include local input for certain data categories
Interface	- Welcome Page and tutorial

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Enhancements	- Short and informal explanation videos - Visual design choices that are relevant to the topic - Filters and search options - Create or upload polygons to analyze a specific area
User Accessibility	- Multilingual options - Offline data access with PDF printable summaries - User accessibility tool

Optional Enhancements (Future Development)	- Resilience solutions feature - Public suggestions and comments are available to everyone
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VI. Conclusion

Through the comparisons among these tools, the recommendations above are intended to help improve the Baldwin Hills URI Tool for future use. After enhancing the tools' prototype to be more accessible and interactive for the public, it will be shared with communities within the BHUWC territory through workshops. The goal is for residents to use this tool to identify areas in the community that should be prioritized and propose resilience projects that the Conservancy may then fund, using Prop 4 funding. This tool will bridge the gap and engage the local communities in planning future projects and discovering which areas are more vulnerable to climate change. The BHUWC will use community input to structure the grants it makes available using Prop 4 funding, and will set its priorities for open space protection and management according to the community's most urgent needs. In the long term, community members will be able to see at least some of the projects they propose funded and implemented in their neighborhoods.

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